

MTH166

Lecture-16

Method of Undeterminant Coeff.

Topic:

Solution of Non-Homogeneous LDE with Constant coefficients.

Learning Outcomes:

To use Method of Undeterminant coefficients to solve Non-Homogeneous LDE with constant coefficients.

Problem 1. Find the general solution of: $4y'' - y = e^{3x}$

Solution: The given equation is:

$$4y'' - y = e^{3x} \quad (1)$$

$$\text{S.F. : } (4D^2 - 1)y = e^{3x} \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (4D^2 - 1) \text{ and } r(x) = e^{3x}$$

To find Complimentary Function (C.F.):

$$\text{A.E. : } f(D) = 0 \quad \Rightarrow (4D^2 - 1) = 0 \quad \Rightarrow D^2 = \frac{1}{4}$$

$$\Rightarrow D = \frac{1}{2}, -\frac{1}{2} \quad (\text{real and unequal roots})$$

$\therefore$ Complimentary function is given by:

$$\Rightarrow y_c = c_1 e^{\frac{1}{2}x} + c_2 e^{-\frac{1}{2}x}$$

To find Particular Integral (P.I.):

Since $r(x) = e^{3x}$, So, let trial solution be: $y_p = ae^{3x}$

$$\Rightarrow y_p' = 3ae^{3x} \quad \Rightarrow y_p'' = 9ae^{3x}$$

Since y_p is a solution of equation (1)

$$\text{So, } 4y_p'' - y_p = e^{3x}$$

$$\Rightarrow 4(9ae^{3x}) - ae^{3x} = e^{3x} \quad \Rightarrow 35ae^{3x} = e^{3x} \quad \Rightarrow 35a = 1 \quad \Rightarrow a = \frac{1}{35}$$

$$\therefore y_p = \frac{1}{35} e^{3x}$$

General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = \left(c_1 e^{\frac{1}{2}x} + c_2 e^{-\frac{1}{2}x} \right) + \frac{1}{35} e^{3x} \quad \textbf{Answer.}$$

Problem 2. Find the general solution of: $y'' + y' - 6y = e^{2x}$

Solution: The given equation is:

$$y'' + y' - 6y = e^{2x} \quad (1)$$

S.F. : $(D^2 + D - 6)y = e^{2x}$ where $D \equiv \frac{d}{dx}$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 + D - 6) \text{ and } r(x) = e^{2x}$$

To find Complimentary Function (C.F.):

$$\text{A.E. : } f(D) = 0 \quad \Rightarrow (D^2 + D - 6) = 0 \quad \Rightarrow (D - 2)(D + 3) = 0$$

$$\Rightarrow D = 2, -3 \quad (\text{real and unequal roots})$$

$$\text{Let } m_1 = 2 \text{ and } m_2 = -3$$

$\therefore$ Complimentary function is given by:

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$\Rightarrow y_c = c_1 e^{2x} + c_2 e^{-3x}$$

To find Particular Integral (P.I.):

Since $r(x) = e^{2x}$, So, let trial solution be: $y_p = axe^{2x}$

$$\Rightarrow y_p' = a[x(2e^{2x}) + e^{2x}(1)] = 2axe^{2x} + ae^{2x}$$

$$\Rightarrow y_p'' = 2a[x(2e^{2x}) + e^{2x}(1)] + 2ae^{2x} = 4axe^{2x} + 4ae^{2x}$$

Since y_p is a solution of equation (1)

$$\text{So, } y_p'' + y_p' - 6y_p = e^{2x}$$

$$\Rightarrow (4axe^{2x} + 4ae^{2x}) + (2axe^{2x} + ae^{2x}) - 6(axe^{2x}) = e^{2x}$$

$$\Rightarrow 5ae^{2x} = e^{2x} \quad \Rightarrow 5a = 1 \quad \Rightarrow a = \frac{1}{5} \quad \therefore y_p = \frac{1}{5}xe^{2x}$$

General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1e^{2x} + c_2e^{-3x}) + \frac{1}{5}xe^{2x} \quad \textbf{Answer.}$$

Problem 3. Find the general solution of: $y'' - 6y' + 9y = 14e^{3x}$

Solution: The given equation is:

$$y'' - 6y' + 9y = 14e^{3x} \quad (1)$$

S.F. : $(D^2 - 6D + 9)y = 14e^{3x}$ where $D \equiv \frac{d}{dx}$

$$\Rightarrow f(D)y = r(x) \text{ where } f(D) = (D^2 - 6D + 9) \text{ and } r(x) = 14e^{3x}$$

To find Complimentary Function (C.F.):

$$\text{A.E. : } f(D) = 0 \Rightarrow (D^2 - 6D + 9) = 0 \Rightarrow (D - 3)(D - 3) = 0$$

$$\Rightarrow D = 3, 3 \quad (\text{real and equal roots})$$

$$\text{Let } m_1 = 3 \text{ and } m_2 = 3$$

$\therefore$ Complimentary function is given by:

$$y_c = (c_1 + c_2x)e^{m_2x}$$

$$\Rightarrow y_c = (c_1 + c_2x)e^{3x} = c_1 e^{3x} + c_2x e^{3x}$$

To find Particular Integral (P.I.):

Since $r(x) = e^{3x}$, So, let trial solution be: $y_p = ax^2e^{3x}$

$$\Rightarrow y_p' = a[x^2(3e^{3x}) + e^{3x}(2x)] = 3ax^2e^{3x} + 2axe^{3x}$$

$$\Rightarrow y_p'' = 3a[x^2(3e^{3x}) + e^{3x}(2x)] + 2a[e^{3x}(1) + x(3e^{3x})] = 9ax^2e^{3x} + 12axe^{3x} + 2ae^{3x}$$

Since y_p is a solution of equation (1)

$$\text{So, } y_p'' - 6y_p' + 9y_p = 14e^{3x}$$

$$\Rightarrow (9ax^2e^{3x} + 12axe^{3x} + 2ae^{3x}) - 6(3ax^2e^{3x} + 2axe^{3x}) + 9(ax^2e^{3x}) = 14e^{3x}$$

$$\Rightarrow 2ae^{3x} = 14e^{3x} \Rightarrow 2a = 14 \Rightarrow a = 7 \quad \therefore y_p = 7x^2e^{2x}$$

General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = c_1 e^{3x} + c_2 x e^{3x} + 7x^2 e^{2x}$$

Answer.

Polling Quiz

For $y'' - 9y = e^{3x}$, the assumed trial solution will be:

(A) $y_p = ae^{3x}$

(B) $y_p = axe^{3x}$

(C) $y_p = ax^2e^{3x}$

Problem 1. Find the general solution of: $y'' - 3y' - 10y = x^2 + 1$

Solution: The given equation is:

$$y'' - 3y' - 10y = x^2 + 1 \quad (1)$$

$$\text{S.F. : } (D^2 - 3D - 10)y = x^2 + 1 \quad \text{where } D \equiv \frac{d}{dx}$$

$$\Rightarrow f(D)y = r(x) \quad \text{where } f(D) = (D^2 - 3D - 10) \quad \text{and} \quad r(x) = x^2 + 1$$

To find Complimentary Function (C.F.):

$$\text{A.E. : } f(D) = 0 \quad \Rightarrow (D^2 - 3D - 10) = 0 \quad \Rightarrow (D - 5)(D + 2) = 0$$

$$\Rightarrow D = 5, -2 \quad (\text{real and unequal roots})$$

$\therefore$ Complimentary function is given by:

$$\Rightarrow y_c = c_1 e^{5x} + c_2 e^{-2x}$$

To find Particular Integral (P.I.):

Since $r(x) = x^2 + 1$, So, let trial solution be: $y_p = ax^2 + bx + c$

$$\Rightarrow y_p' = 2ax + b \quad \Rightarrow y_p'' = 2a$$

Since y_p is a solution of equation (1)

$$\text{So, } y_p'' - 3y_p' - 10y_p = x^2 + 1$$

$$\Rightarrow 2a - 3(2ax + b) - 10(ax^2 + bx + c) = x^2 + 1$$

$$\Rightarrow -10ax^2 - (6a + 10b)x + (2a - 3b - 10c) = x^2 + 1$$

Comparing the like coefficients:

$$\text{Coeff. of } x^2: -10a = 1 \quad \Rightarrow a = -\frac{1}{10}$$

$$\text{Coeff. of } x: -(6a + 10b) = 0 \quad \Rightarrow 10b = -6a \quad \Rightarrow b = -\frac{6}{100}$$

$$\text{Constant terms: } (2a - 3b - 10c) = 1 \Rightarrow 10c = 2a - 3b - 1$$

$$\Rightarrow c = \frac{1}{10}(2a - 3b - 1) = \frac{1}{10}\left(2\left(\frac{1}{10}\right) - 3\left(\frac{-6}{100}\right) - 1\right) = -\frac{62}{1000}$$

Putting back values of a, b, c in $y_p = ax^2 + bx + c$

$$y_p = \frac{1}{10}x^2 - \frac{6}{100}x - \frac{62}{1000}$$

General solution is given by: $y = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } y = y_c + y_p$$

$$\Rightarrow y = (c_1 e^{5x} + c_2 e^{-2x}) + \frac{1}{10}x^2 - \frac{6}{100}x - \frac{62}{1000} \quad \textbf{Answer.}$$

Polling Quiz

For $y'' - 3y' - 10y = x^3 + 1$, the assumed trial solution will be:

(A) $y_p = ax^3 + bx^2 + cx + d$

(B) $y_p = a + bx + cx^2 + dx^3$

(C) Both A and B

(D) None of these.

